

LINYAN WAN

Fermilab, PO Box 500, Batavia, IL 60510
lwan@fnal.gov

EMPLOYMENT

Wilson Fellow Fermilab, Batavia, IL, USA	July 2023 - present
Postdoctoral Associate Department of Physics, Boston University, Boston, MA, USA	May 2019 - July 2023

EDUCATION

Ph.D. in Experimental Particle Physics Department of Engineering Physics, Tsinghua University, Beijing, China	Sep. 2013 - Jan. 2019
B.S. in Engineering Physics Department of Engineering Physics, Tsinghua University, Beijing, China	Sep. 2009 - Jul. 2013

RESEARCH EXPERIENCE

SBND experiment • Serving as the SBND physics coordinator. • Leading the study on neutron-induced blip reconstruction [1]. • Serving as the campaign manager for the 2nd generation of SBND data production and processing, enabling cross-section and beyond-standard model physics analyses. • Served as the calibration group co-convenor, led and successfully completed the first calibration campaign for SBND. • Led the diagnostic measurement and monitoring for the LArTPC high-voltage system at SBND, serving as the High-Voltage expert.	Jul. 2023 - present
DUNE experiment • Leading the sensitivity study for neutron-antineutron oscillation search. • Leading the detector surrogate model for fast detector uncertainty evaluation (DetSuM). • Collaborating on the transformerCVN based neutrino reconstruction for atmospheric neutrino oscillation analysis.	Jul. 2023 - present
Super-Kamiokande experiment • Pioneering a novel inclusive search for proton decay, ongoing study [2]. • Led a novel search for boosted dark matter using recoil protons, yielding the world's most stringent limit on sub-GeV dark matter-hadron coupling [10]. • Led $n \rightarrow \bar{n}$ oscillation search, yielding the world's most stringent limit [11]. • Conducted the DSNB search with crucial contribution on neutron tagging development [12]. • Collaborated on the atmospheric neutrino oscillation analysis with fundamental contributions on simulation, data selection, and systematic uncertainty constrain [4][8]. • Led the measurement of atmospheric neutrino NCQE cross-section [15][9]. • Conducted the solar neutrino energy reconstruction and calibration, with major contribution on PMT gain correction, quantum efficiency estimation, and multi-source calibration [7]. • Conducted the water quality calibration and monitoring with LEDs. • Served as the outer detector high voltage expert.	Dec. 2013 - present
EMPHATIC experiment	Nov. 2019 - present

- Served as the run coordinator for Phase 1b.
- Led the geometry construction and calibration for software development, with major contribution to channel mapping and reconstruction.
- Led the design and development of an aerogel ring imaging Cherenkov detector (ARICH).
- Mentored multiple student projects.

T2K experiment

Jan. 2021 - Jan. 2024

- Served as the reviewer for the T2K-SK joint-fit neutrino oscillation analysis [5].

Jinping Neutrino Experiment

Sep. 2013 - Jan. 2019

- Constructed the energy-response model for liquid scintillator detectors [17].
- Conducted the sensitivity estimation for the geoneutrino proposal [18].
- Conducted the solar neutrino simulation, with studies on CNO neutrino flux, day-night asymmetry, and metallicity [19].
- Led the development of the simulation software package [19].

SEMINARS AND COLLOQUIA

- 2025.09.25 Seminar "Atmospheric Neutrinos: History, and Future", Physics & Astronomy Colloquium, Louisiana State University, Baton Rouge (USA)
- 2025.07.02 Colloquium "Solar Neutrinos and Atmospheric Neutrinos", Neutrino University, Fermilab (USA)
- 2025.01.09 Seminar "Atmospheric Neutrino and Baryon Number Violation: Opportunities in LArTPC", Particle Physics Seminar, University of California, San Diego (USA)
- 2025.01.08 Seminar "Atmospheric Neutrino Opportunities in LArTPC", Particle Physics Seminar, University of California, Irvine (USA)
- 2024.11.01 Colloquium "Atmospheric Neutrinos", Physics Division Colloquium, Argonne National Lab (USA)
- 2023.10.05 Seminar "Atmospheric Neutrinos", Neutrino Seminar Series, Fermilab (USA)
- 2023.08.09 Colloquium "Solar Neutrinos and Atmospheric Neutrinos", Neutrino University, Fermilab (USA)
- 2023.03.08 Seminar "Boosted Dark Matter Search at Super-Kamiokande", HEP Group Seminar, University of Harvard (USA)
- 2023.03.06 Colloquium "Atmospheric Neutrinos: Neutrino Oscillation, and Beyond", Physics Department, Florida State University (USA)
- 2023.02.27 Seminar "Boosted Dark Matter Search at Super-Kamiokande", Physics Department, University of Toronto (via Zoom)
- 2023.02.02 Seminar "Boosted Dark Matter Search at Super-Kamiokande", Physics Department, Indiana University (USA)
- 2022.07.22 Seminar "Baryon Number Violation at Super-Kamiokande", HEP Seminar, Tsinghua University (via Zoom)
- 2022.06.15 Seminar "Astrophysics at Super-Kamiokande", EPD Seminar, Institute of High Energy Physics (via Zoom)
- 2022.04.21 Special Colloquium "Physics at Super-Kamiokande", TRIUMF (via Zoom)
- 2020.02.27 Seminar "Searching for Diffused Supernova Neutrino Background at Super-Kamiokande", Department of Physics, Boston University (USA)
- 2018.09.27 Seminar "Supernova Relic Neutrino Search and Related Topics at Super-Kamiokande", Center for High Energy Physics, Peking University (China)

CONFERENCE TALKS AND POSTERS

Invited talks

- 2026.06.22 "First Physics Results from the Short-Baseline Near Detector", NEUTRINO 2026, University of California, Irvine (USA)
- 2025.07.18 "Status of the Short-Baseline Near Detector Experiment", 58th Annual Users and Affiliates Meeting 2025: Forward to Discory through Innovation, Fermilab (USA)
- 2025.01.13 "Search for Neutron-Antineutron Transition in Large Neutrino Detectors", Baryon Number Violation: From Nuclear Matrix Elements to BSM Physics, Institute for Nuclear Theory (USA)
- 2024.05.15 "Overview of Nucleon Decay", CoSSURF 2024, South Dakota Mines (USA)
- 2024.04.05 "Astrophysical Neutrinos with Large Neutrino Experiments", APS April Meeting 2024, American Physical Society (USA)
- 2022.06.02 "New Results on Atmospheric Neutrinos at Super-Kamiokande", NEUTRINO 2022, Institute for Basic Science (via Zoom)
- 2020.08.03 "Searching for Neutron Antineutron Oscillation at Super-Kamiokande & Prospect for SK-Gd and HK", ACFI workshop, University of Massachusetts Amherst (via Zoom)
- 2019.11.08 "Super-Kamiokande Status and Prospect for SK-Gd", NNN19, Universidad de Medellin (Colombia)
- 2017.07.15 "Geoneutrino detection at Jinping", Workshop of Jinping Neutrino Experiment, Tsinghua University (China)
- 2016.10.26 "Proposal: Low-energy Neutrino Research at Jinping", International Workshop: Neutrino Research and Thermal Evolution of the Earth, Tohoku University (Japan)

Posters and Contributed Talks

- 2024.05.16 "Diffuse Supernova Neutrino Background Search", CoSSURF, South Dakota Mines (USA)
- 2022.06.02 Poster "Boosted Dark Matter Search with Hadrons at Super-Kamiokande", NEUTRINO 2022, Institute for Basic Science (via Zoom)
- 2022.05.12 Talk "Nucleon Decay Studies at Super-Kamiokande", CoSSURF, South Dakota Mines (USA)
- 2020.06.24 Poster "Neutron-antineutron oscillation search at Super-Kamiokande", NEUTRINO 2020, Fermilab (via virtual reality)
- 2018.07.06 Talk "Status of the Supernova Relic Neutrino Search and Atmospheric Neutrino Neutral-Current Quasi-Elastic Interactions Measurement at Super-Kamiokande", ICHEP 2018, Institute for Basic Science (South Korea)
- 2018.06.20 Talk "Supernova Relic Neutrino Search and Atmospheric Neutrino NCQE at Super-Kamiokande", Chinese Physics Society Meeting, Shanghai Jiao Tong University (China)
- 2018.06.14 Talk "Simulation and sensitivity studies for solar neutrinos at Jinping", 5th International Solar Neutrino Conference, Dresden University of Technology (Germany)
- 2018.06.07 Poster "First Measurement of Atmospheric Neutrino Neutral-Current Quasi-Elastic Interactions for Supernova Relic Neutrino Search at Super-Kamiokande", NEUTRINO 2018, Max-Planck-Institut für Kernphysik (Germany)
- 2016.07.04 Poster "Neutrino telescopes with the deepest overburden: The Jinping neutrino experiment", NEUTRINO 2016, Imperial College London (UK)

SELECTED PUBLICATIONS

1. Neutron Reconstruction via Blips in Liquid Argon Time Projection Chambers
(*Corresponding author*) Miguel Hernandez Morquecho, Bryce Littlejohn, Paola Sala, *Linyan Wan*, Phys. Rev. D **114**, 012013 (2026).
2. Baryon number violation: from nuclear matrix elements to BSM physics
Leah J. Broussard et al, J. Phys. G **52** 8, 083001 (2025).
3. The track-length extension fitting algorithm for energy measurement of interacting particles in liquid argon TPCs and its performance with ProtoDUNE-SP data
DUNE Collaboration, JINST, **20** 02, P02021 (2025).

4. Measurement of neutron production in atmospheric neutrino interactions at Super-Kamiokande
Super-Kamiokande Collaboration, Phys. Rev. D, **112** 1, 012004 (2025).
5. First joint oscillation analysis of Super-Kamiokande atmospheric and T2K accelerator neutrino data
T2K and Super-Kamiokande Collaborations, Phys. Rev. Lett., **134** 1, 011801 (2025).
6. Scintillation light in SBND: simulation, reconstruction, and expected performance of the photon detection system
SBND Collaboration, Eur.Phys.J.C, **84** 10, 1046 (2024).
7. Solar neutrino measurements using the full data period of Super-Kamiokande-IV
Super-Kamiokande Collaboration, Phys. Rev. D, **109** 9, 092001 (2024).
8. Atmospheric neutrino oscillation analysis with neutron tagging and an expanded fiducial volume in Super-Kamiokande I–V
Super-Kamiokande Collaboration, Phys. Rev. D, **109**, 072014 (2024).
9. Measurement of the neutrino-oxygen neutral-current quasielastic cross section using atmospheric neutrinos in the SK-Gd experiment
Super-Kamiokande Collaboration, Phys. Rev. D, **109** 1, L011101 (2024).
10. Search for Cosmic-ray Boosted Sub-GeV Dark Matter using Recoil Protons at Super-Kamiokande
(*Corresponding author*) Super-Kamiokande Collaboration, Phys. Rev. Lett., **130**, 031802 (2023).
11. Neutron-antineutron oscillation search using a 0.37 megaton-years exposure of Super-Kamiokande
(*Corresponding author*) Super-Kamiokande Collaboration, Phys. Rev. D, **103**, 012008 (2021).
12. Diffuse supernova neutrino background search at Super-Kamiokande
Super-Kamiokande Collaboration, Phys. Rev. D, **104**, 122002 (2021).
13. Search for solar electron anti-neutrinos due to spin-flavor precession in the Sun with Super-Kamiokande-IV
Super-Kamiokande Collaboration, Astropart.Phys. **139**, 102702 (2022).
14. Muon Flux Measurement at China Jinping Underground Laboratory
JNE Collaboration, Chin. Phys. C, **45**, 025001 (2021).
15. Measurement of the neutrino-oxygen neutral-current quasielastic cross section using atmospheric neutrinos at Super-Kamiokande.
Linyan Wan, et al., Phys. Rev. D, **99**, 032005 (2019).
16. Sensitivity of Super-Kamiokande with Gadolinium to Low Energy Anti-neutrinos from Pre-supernova Emission
Super-Kamiokande Collaboration, Astrophys. J. **885**, 133 (2019).
17. An efficient energy response model for liquid scintillator detectors.
Logan Lebanowski, *Linyan Wan*, et al., Nucl. Instrum. Meth. A. **890**, 133-141 (2018).
18. Geoneutrinos at Jinping: Flux prediction and oscillation analysis.
Linyan Wan, Ghulam Hussain, Zhe Wang, Shaomin Chen, Phys. Rev. D, **95**, 053001 (2017).
19. Physics prospects of the Jinping neutrino experiment.
Jinping Collaboration, Chin. Phys. C, **41**, 023002 (2017).

SERVICE WORK

- Organizer for SBND 2025 Analysis Workshop and SBND 2024 Calibration Workshop at Fermilab
- Local organization committee for CU**iP* 2025 and NuFact 2024
- Referee for Physical Review Letters, Physical Review X, European Physical Journal C, and Chinese Physics C

MENTORING AND OUTREACHES

- 2026 Mentor of Kevin Wu, a MetCalf student at Fermilab, on antineutron annihilation researches.
- 2024 Mentor of Justin Wheeler, a SULI student at Fermilab, on neutron-antineutron transition search at DUNE.
- 2024 Author for *Atmospheric Neutrinos Revisited*, Viewpoint, Physics 16, 212 (2023).

- 2019-2023 Mentor of Boston University undergraduate and graduate students, on simulation, sensitivity estimation, and data analysis for EMPHATIC experiment
- 2017 Invited speech to undergraduates at Miyagi University of Education, on a virtual tour to Super-Kamiokande, invited by Prof. Yoshiyuki Fukuda
- 2015 Translator for *Quantum Field Theory*. Science Press. LLC (2015), 978-7030458971 (English to Chinese).